1 Lesson 11-11-26

1.1 I/O mapping vs memory mapping

bus transaction is characterized by the address generated by the bus master (i.e. the processor core). The Address Bus of the On-Chip Bus has a certain number of bits, where 32 bits is a typical value, for a maximum addressable space of 4GB. The amount of memory in the system is generally less than the maximum addressable space: a system may have only 128MB out of 4GB of addressable space.

Despite this, the master can generate any address: also addresses where there is no memory. This can be exploited to access the peripheral registers

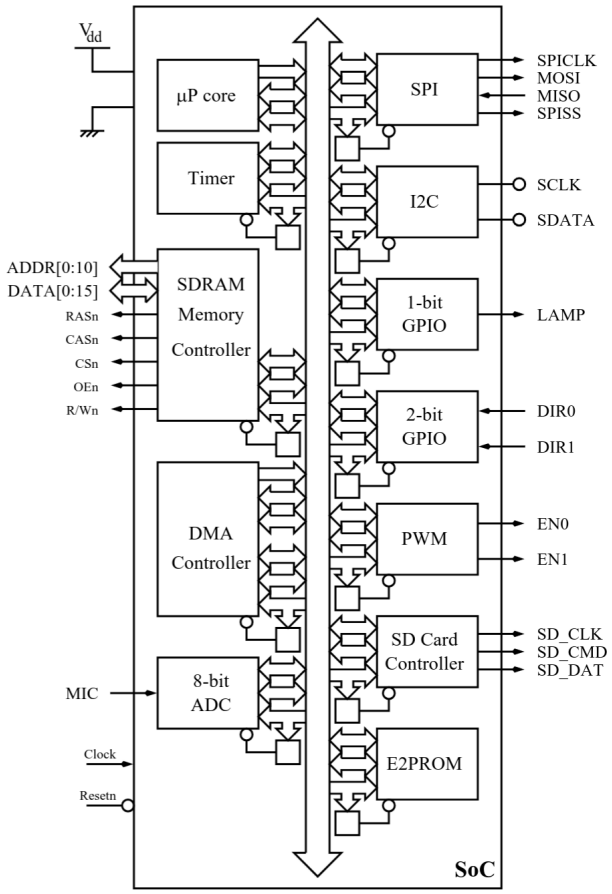
So, we know the two big categories of peripheral mapping in a processor's addressing space:

- **I/O mapping**: registers in a peripheral are assigned a contiguous segment of addresses within a dedicated addressin space, called the *I/O space*. This doesn't exploit our previous property. Peripheral addresses can overlap memory addresses: what's important is that a peripheral address range cannot overlap the range of another peripheral. To distinguish between memory and peripheral (I/O) operation, a special signal of the On-Chip Bus is used. The addressable memory space for I/O operations is often just a subset of the available addressable space (only some address wires are used, rather than all of them as when targeting a memory).
- **Memory mapping**: registers in a peripheral are assigned a contiguous segment of addresses, similar to I/O mapping. The difference is that peripheral addresses should not overlap memory addresses: no need to distinguish!
  - A peripheral address range cannot overlap the range of another peripheral or memory
  - A memory address range cannot overlap the range of another memory or peripheral

  Peripheral addresses should not be cached, if a cache is present. Registers are modified by the peripheral, as well as by the processor A special instruction might be used to bypass the cache (Nios II). The GNU compiler collection, for instance, offers specific flags, and the C language itself has `volatile`.

1.2 SoCs

We can now talk about **SoC**s (which stands for *System on Chip*).



The processor core has no external pins on the SoC: the chip includes DMA Controller, E2PROM,timer and other peripherals. Processor core and DMA are the only masters on the On-Chip Bus: DMA is also a slave (with two separate ports on the bus).

Slave interfaces include address decoders, which drive the peripheral Chip Select signals of the respective bus interfaces.

Clock and Reset go to all blocks in the system.

We said that the processor is a master on the On-Chip Bus, peripheral is a slave (except DMAs). This means the master can generate addresses, and it determines the kind of operation (read, write, ...).

How can a peripheral request attention to the processor? 3 main ways:

- **Polling** (high latency, low performance), might loose some data, as its not real time;
- **Interrupt** (medium latency, medium performance), needs hardware in the processor (which usually does exist);
- **DMA** *Direct Memory Access* (small latency, high performance), we need a special peripheral with master interface on the On-Chip Bus.